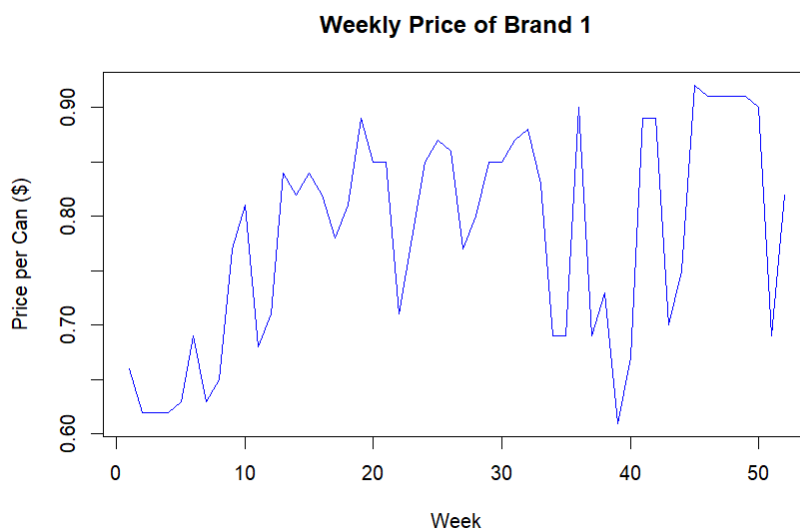
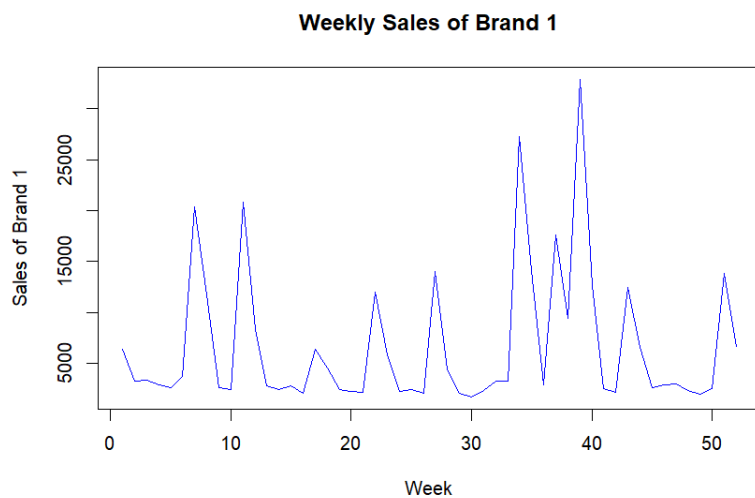


Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable *SAL1* = unit sales of brand no. 1 canned tuna, and *APR1* = price per can of brand no. 1 tuna (in dollars).

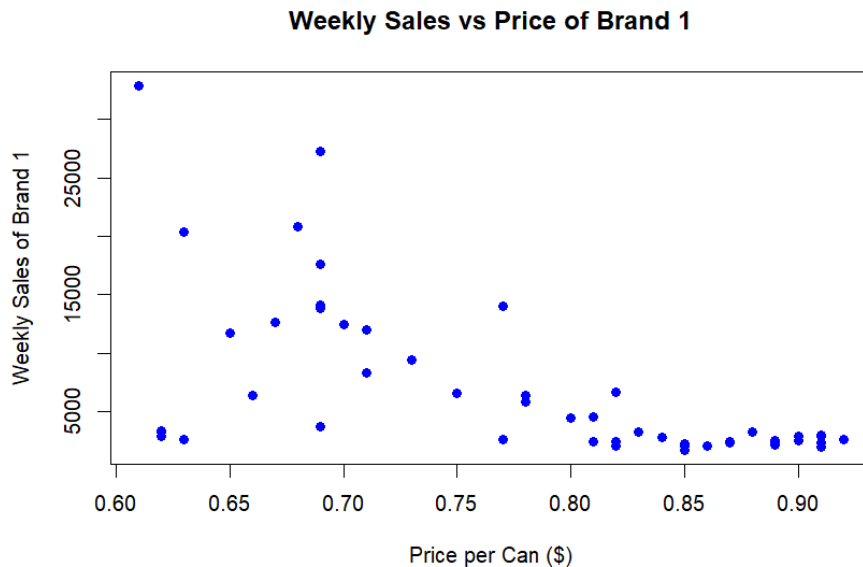
a. Calculate the summary statistics for *SAL1* and *APR1*. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?

	sample means	minimum	maximum	standard deviations
<i>SAL1</i>	6718.712	1762.00	32820.00	6915.083
<i>APR1</i>	0.7825	0.61	0.92	0.098



The weekly sales of brand 1 tuna vary quite a bit. The standard deviation is 6915 units, with weekly sales ranging from 1,762 to 32,820 units. This indicates that sales fluctuate substantially from week to week. In contrast, the weekly price per can shows much less variation. The standard deviation is 0.098 dollars, with prices ranging from \$0.61 to \$0.92. This suggests that the price remained relatively stable throughout the year.

b. Plot the variable *SAL1*(y-axis) against *APR1*(x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?



The scatter plot of weekly sales (*SAL1*) against price per can (*APR1*) shows an inverse relationship: as price increases, sales tend to decrease. This is consistent with what we expect from basic economic theory, where higher prices usually reduce the quantity demanded.

c. Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

The point estimate indicates that for every one cent increase in the price of brand no. 1 tuna, weekly sales decrease by approximately 434.4473 units. The 95% confidence interval for this effect is [-592.2897, -276.6049], suggesting that we are 95% confident that the true decrease in sales for a one cent price increase lies within this range.

d. Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

When the price per can is 70 cents, the expected number of cans sold per week is approximately 10,302.9. The 90% confidence interval for the expected sales is between 8,625 and 11,981 cans. This means we are 90% confident that the true average weekly sales at a price of 70 cents lies within this range.

e. Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 “at the means.” Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

$$E = \frac{\partial SAL1}{\partial PRICE1} \cdot \frac{PRICE1_{mean}}{SAL1_{mean}} = \beta_2 \frac{PRICE1_{mean}}{SAL1_{mean}}$$

The price elasticity of sales at the sample means is estimated as:

$$E = (-434.7743) \times \left(\frac{78.25}{6718.712} \right) = -5.0598$$

This indicates that a 1% increase in the price of canned tuna leads to roughly a 5.06% decrease in weekly sales.

The 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 “at the means is [-6.8981,-3.2215]. Since $|E| > 1$, sales are highly elastic with respect to price, which makes sense because consumers have many substitutes and are sensitive to price changes.

f. Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p-value for the test. What is your conclusion?

The hypotheses are:

$$H_0: E = -3$$

$$H_1: E \neq -3$$

The test statistic is

$$t = \frac{\text{elasticity} - (-3)}{se(\text{elasticity})} \cong -2.250572$$

At the 10% significance level, the critical value is approximately 1.675905. The rejection region is: $|t| > 1.675905$. Since $|-2.250572| > 1.675905$, we reject the null hypothesis. The p-value is approximately 0.02884204, which is smaller than 0.1, so we reject H_0

We reject the null hypothesis. There is sufficient evidence at the 10% significance level to conclude that the elasticity is significantly different from -3.